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GREATER NOIDA, UTTAR PRADESH

AICTE & ICAC APPROVED INTERNSHIP

Full-Stack School Management System

An Internship Presentation Report on SmartEdu

INTERNSHIP DURATION

May 11, 2026 – July 11, 2026 (2 Months)

STUDENT CREDENTIALS

Simmi Tiwari

Enrollment / Roll No.
2410031627

Program & Semester
B.Tech (CSE), Batch 2026-27

HOST ORGANIZATION
Codec Technologies Pvt. Ltd.
Supported by AICTE National Internship Portal



Academic & Candidate Profile

Student: Simmi Tiwari

Roll Number: 2410031627 (Section 3CSE9)

Degree: B.Tech (Computer Science & Engineering)

University: IILM University, Greater Noida

Industry & Program Context



Organization: Codec Technologies Pvt. Ltd.

Role: Full Stack Developer Intern

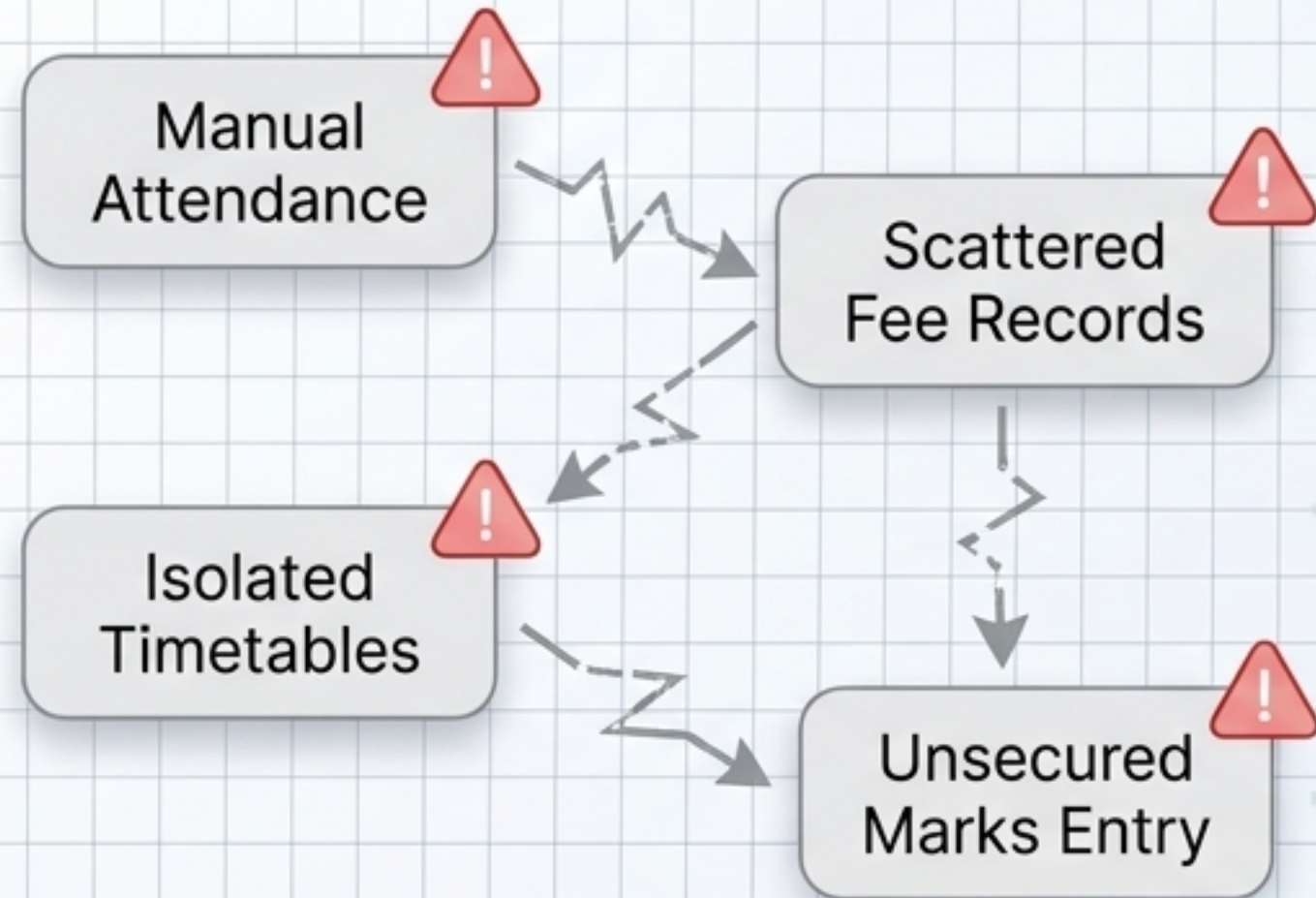
Duration: 2 Months (11/05/2026 – 11/07/2026)

Authority: AICTE & ICAC Approved Internship



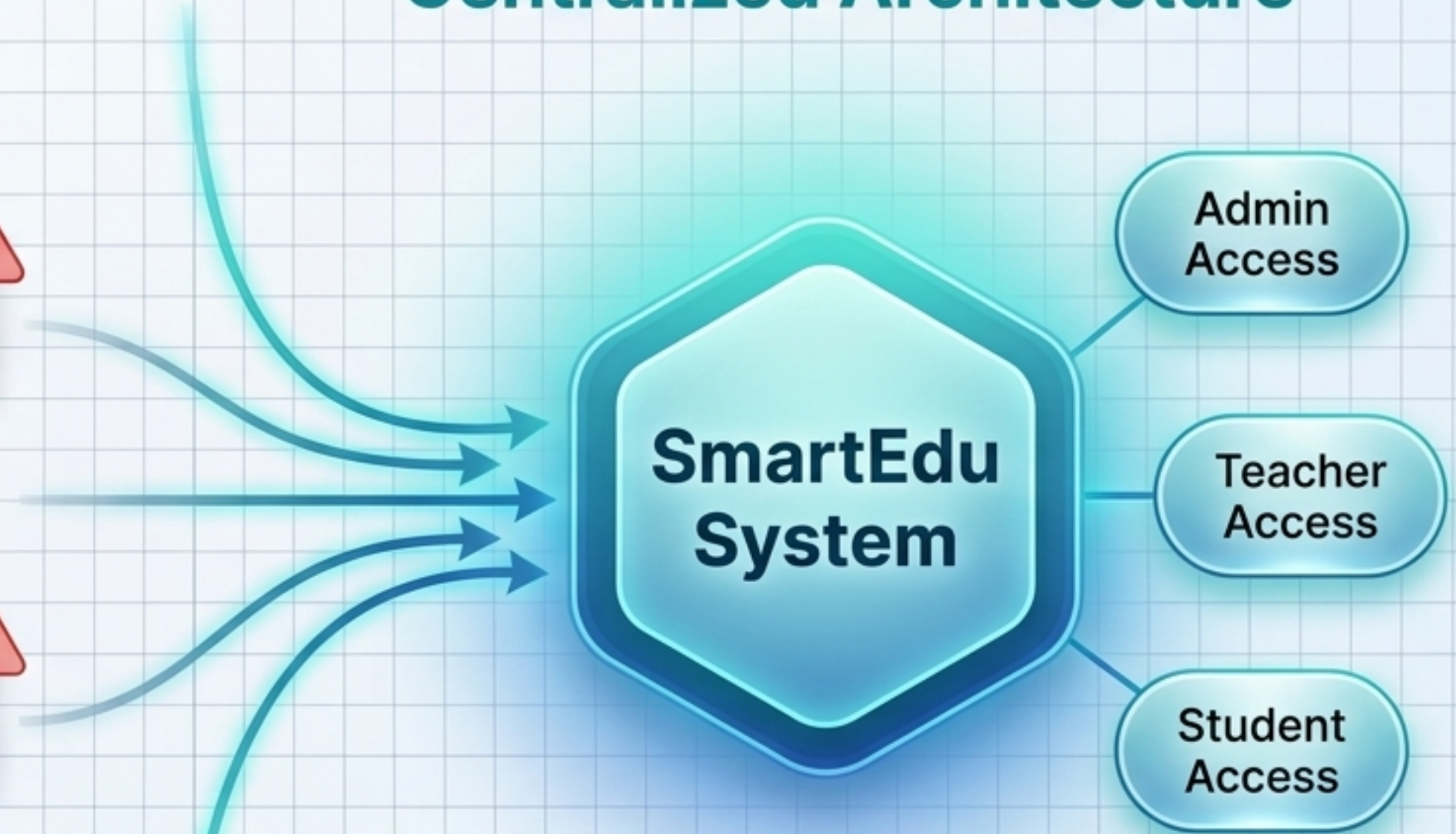
Designed, developed, and deployed SmartEdu, a secure, role-based School Management System integrating front-end interfaces with a PHP/MySQL backend.

The Problem: Isolated Data Silos



Educational institutions struggle with disconnected records, manual processes, and a critical lack of role-based data security.

The Objective: Centralized Architecture

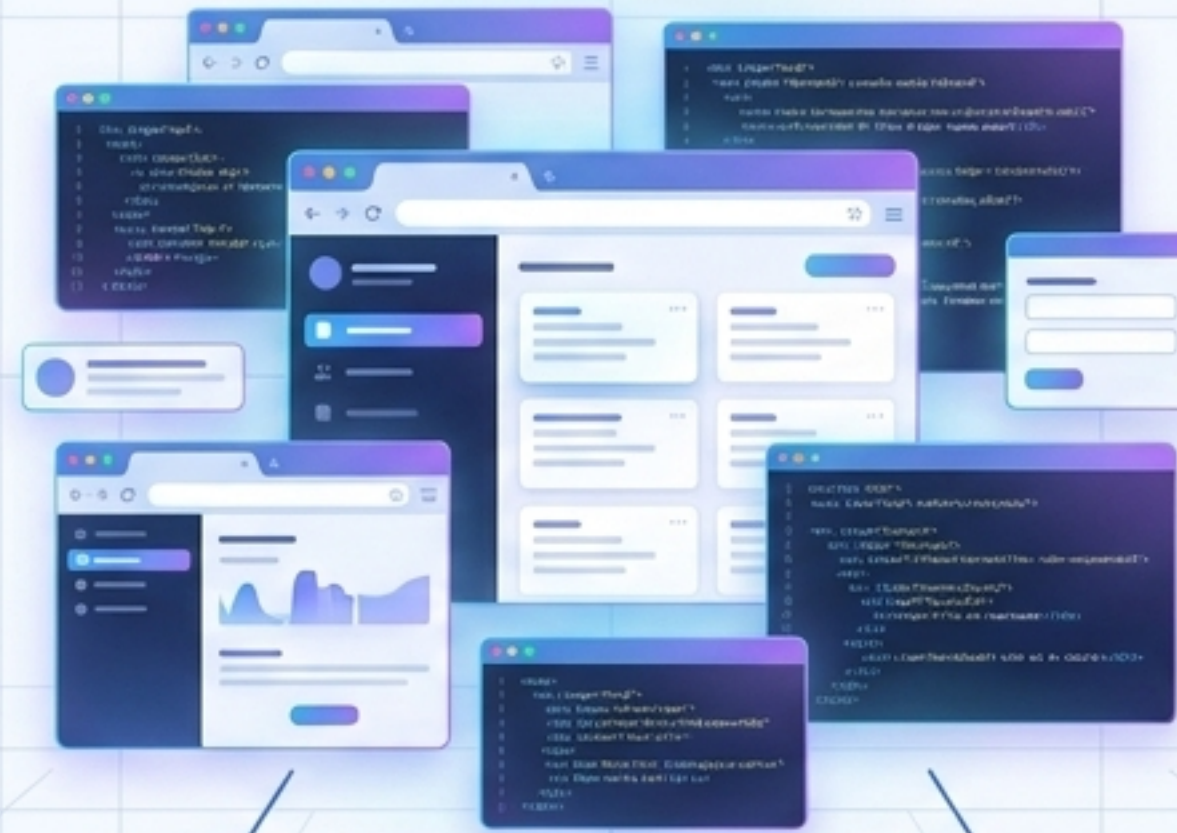


A centralized environment with role-specific routing to ensure data security, operational efficiency, and instantaneous retrieval.

SmartEdu: The Centralized Solution

Responsive Web Interface

Fluid layouts spanning desktop, tablet, and mobile environments.



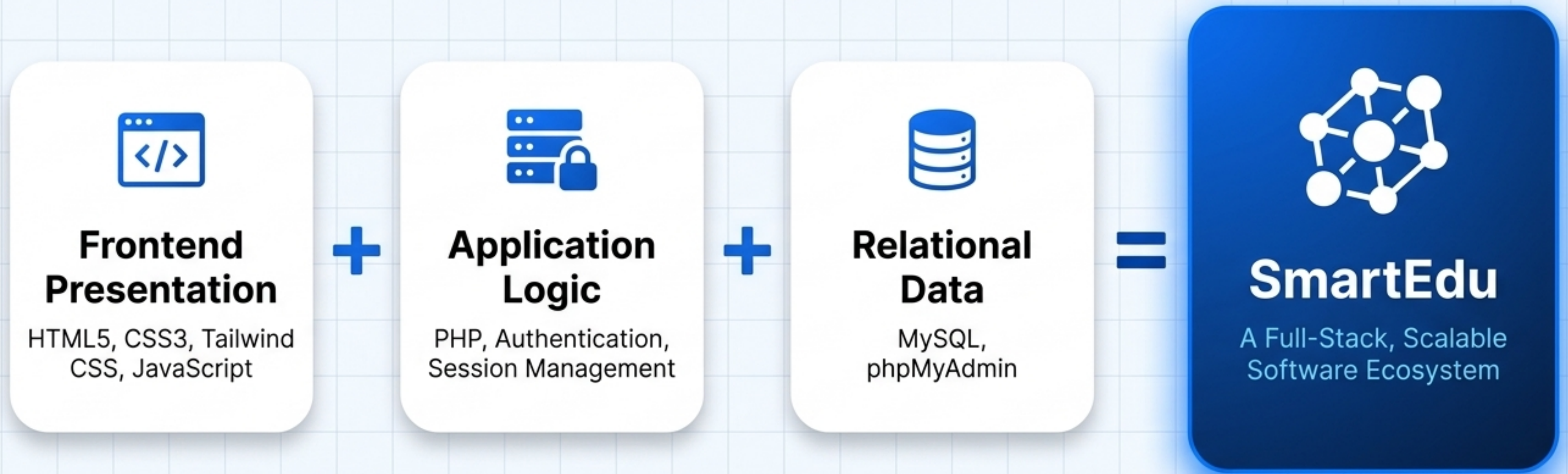
Role-Based Dashboards

Distinct, secure operational views for Administrators, Teachers, and Students.

Real-Time Data Integration

Instantaneous synchronization between the MySQL backend and frontend views.

System Architecture & Tech Stack



Development Environment: Supported by VS Code as the primary IDE and GitHub for version control and collaborative debugging.

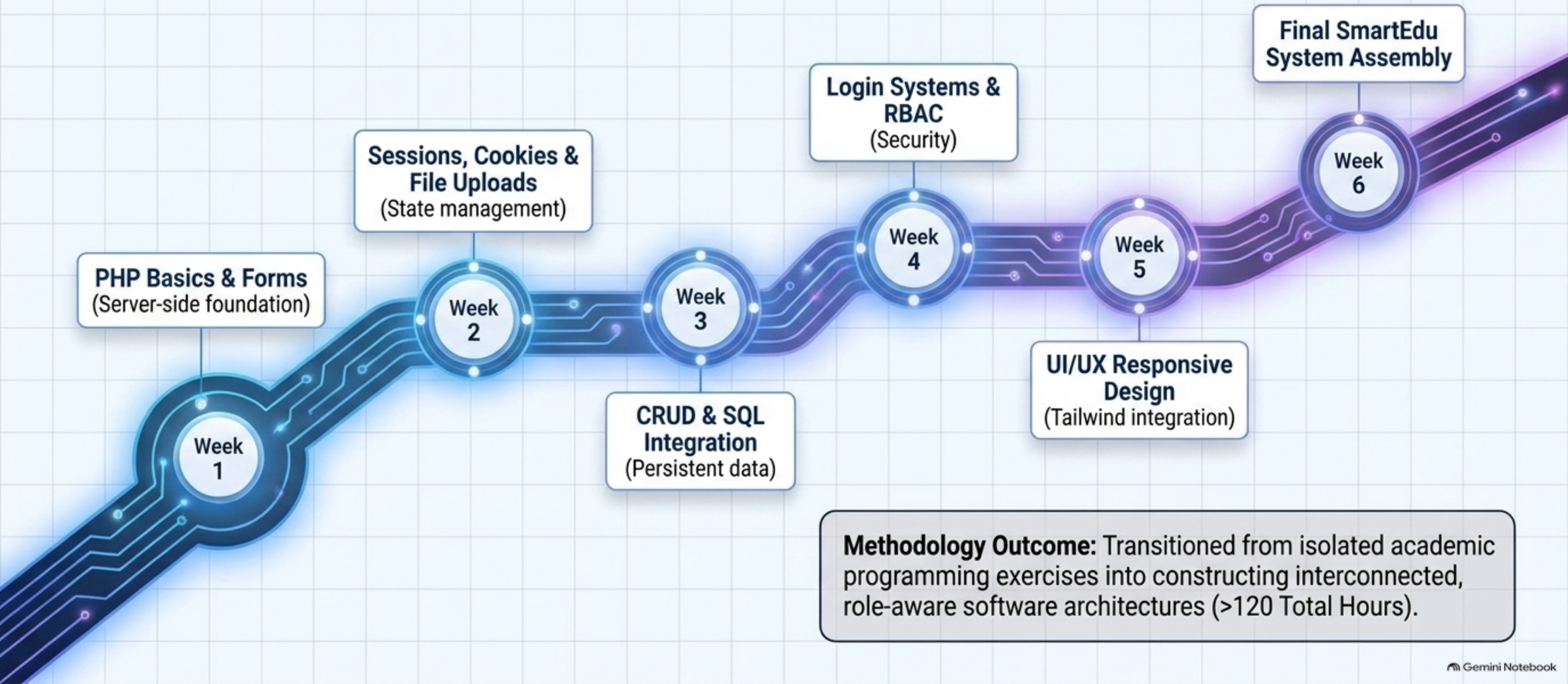
Role-Based Access Control (RBAC) Matrix

	Dashboard Overview	Manage Users	View Timetable	Mark Attendance	Enter Marks	Fee Management
Administrator	✓	✓	✓	—	—	✓
Teacher	✓	—	✓	✓	✓	—
Student	✓	—	✓	—	—	✓

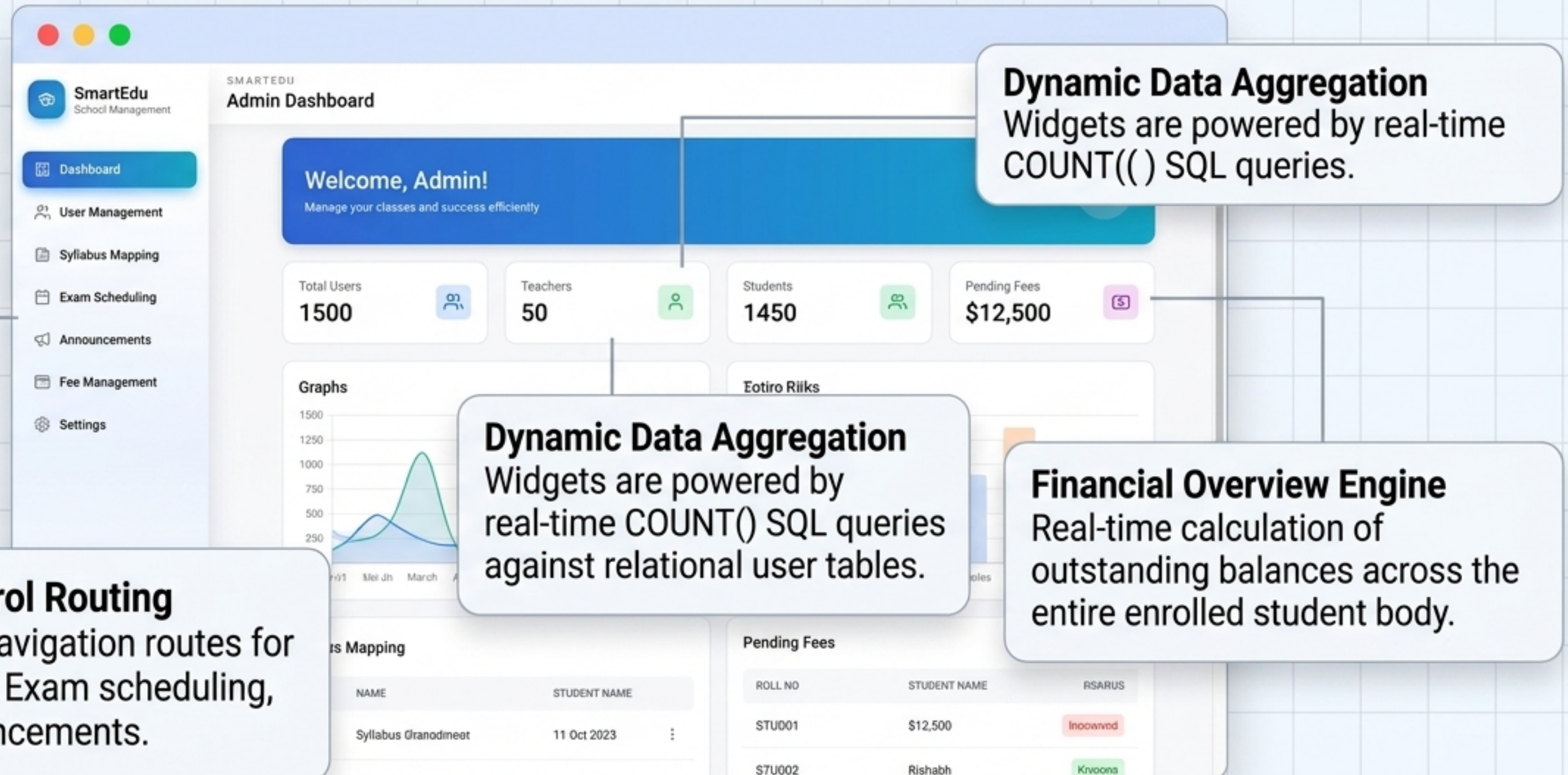


Security Protocol: PHP session variables are strictly enforced at the server level, automatically routing authenticated users exclusively to their designated system portals.

Development Methodology: The 6-Week Sprint



Feature Spotlight: Administrator Operations



Feature Spotlight: Teacher Portal & Academic Engine



SmartEdu
School Management

Dashboard

Classes

Students

Attendance

Marks

Syllabus

Settings

Settings

Teacher Dashboard

Welcome to SmartEdu!
Welcome to your icos welcome to mnvad your dashboard.

Quick Str

SmartEdu
School Management

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Marks Management

Class: Class 10A Subject: Mathematics

Attendance

Rahul Kumar

☒

Present

Priya Sharma

☐

Absent

Aditya Singh

☒

Present

Roll No.	Student Name	Marks (Out of 100)	Total Marks
001	Rahul Kumar	85	
002	Priya Sharma	87	
003	Aditya Singh	25	
004	Rahul Kumar	70	90
005	Priya Kumar	50	89

1

Contextual Filtering: Dropdowns dynamically load specific class and subject rosters from the relational database.

2

Attendance State Management: Intuitive toggle inputs for Present/Absent status updating relational tables in real-time.

3

Grade Book Engine: Secure entry fields mapping numerical scores to specific student IDs and examination periods.

Gemini Notebook

Feature Spotlight: Attendance State Management

The screenshot shows the 'SmartEdu Attendance Management' web application. It features a header with the logo and title, and a user profile in the top right. The main section is titled 'Attendance' and contains two dropdown menus: 'Select Class' and 'Select Date'. Below these is a table with columns for 'Roll No.', 'Student Name', and 'Status'. The table lists six students with their roll numbers and names. Each row has a 'Status' column with two buttons: a green 'Present' button and a red 'Absent' button. Two callout boxes are overlaid on the interface. The first callout, titled 'Targeted Roster Loading', points to the 'Select Class' and 'Select Date' dropdowns. The second callout, titled 'Instant State Translation', points to the 'Present' button for Student Name 3.

Targeted Roster Loading
PHP logic securely fetches only the students assigned to the authenticated teacher's active classes.

Instant State Translation
Intuitive UI toggles translate instantly into boolean status updates within the MySQL attendance tables.

Roll No.	Student Name	Status
31	Student Name 1	☒ Present ☐ Absent
32	Student Name 2	☒ Present ☐ Absent
33	Student Name 3	☒ Present ☐ Absent
34	Student Name 4	☒ Present ☐ Absent
35	Student Name 5	☒ Present ☐ Absent
36	Student Name 6	☒ Present ☐ Absent

Feature Spotlight: Student Dashboard



SmartEdu
School Management

Dashboard

Classes

Students

Attendance

Marks

Syllabus

Settings

Student Portal

Welcome back, Student Name!
Welcome to your iros welcome to mnvag your dashboard.

Quick St...

SmartEdu
School Management

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Marks

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Attendance

30
Points ts

Upcoming Classes

Embeddeds Classoot

Timetable

1 / 7 100% +

Class: Student Name

Class: 10A

Class	Mon	Tue	Wed	Thu
Class 1	Orndint Name	Wnshnday Name	Wednesday Name	Conner
Class 2	Sloafent Name	Wadnerakes	Bartant Name	PGP
Class 3	Sloafent Name	Sloafent Name	Andraent Name	Sedmar
Class 4	Restant Name	Stamionos	Sedant Name	Stalone
Class 5	Wednesday Name	Wdnesday Name	Wednesday Name	PGP
Class 6	Sedant Name	Wadnesday Name	Sedant Name	PGP

1

Session-Locked View: PHP session variables ensure students can strictly view only their own academic data.

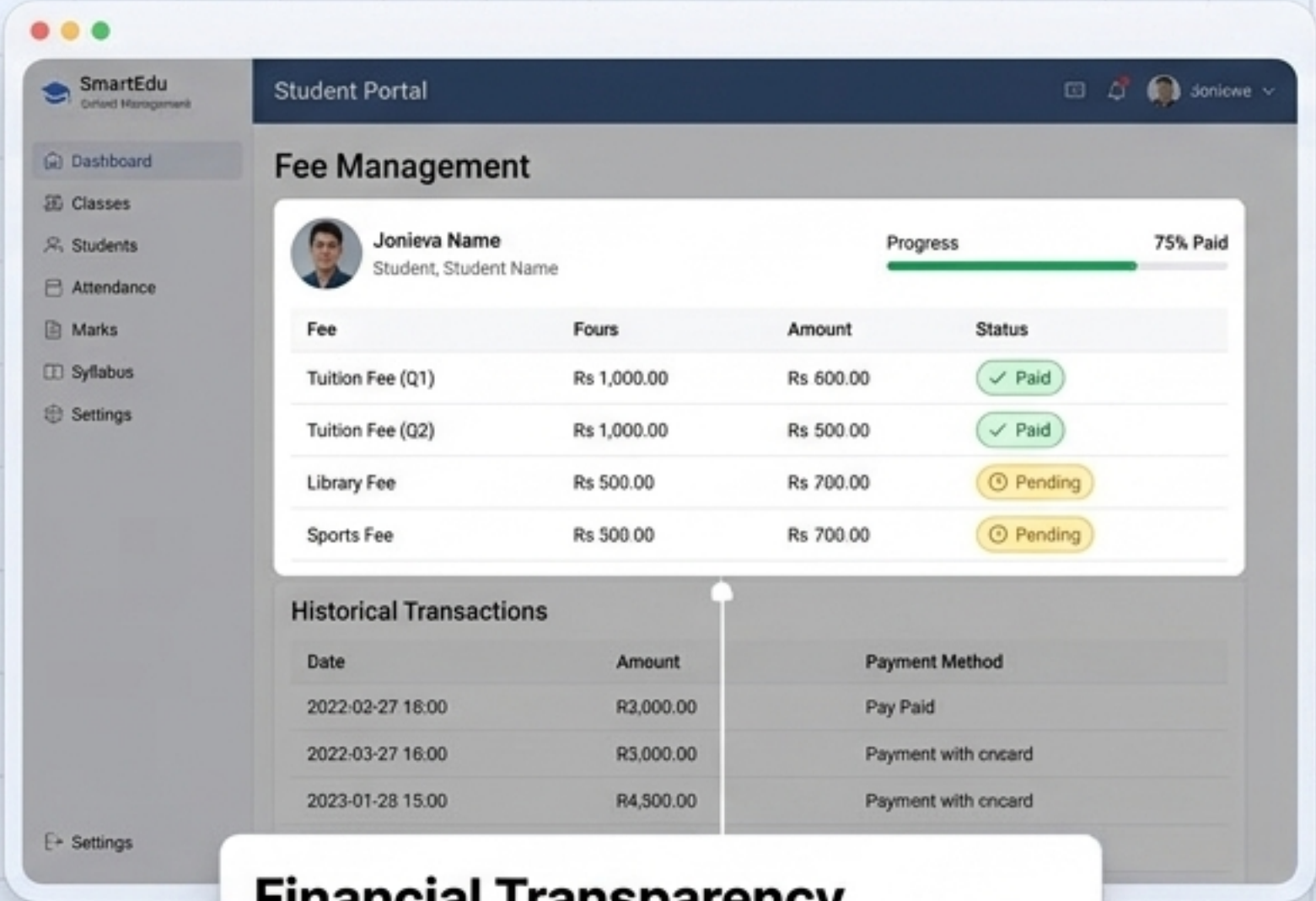
2

Visual Data Representation: Progress bars and color-coded status badges translating raw database numbers into readable attendance metrics.

3

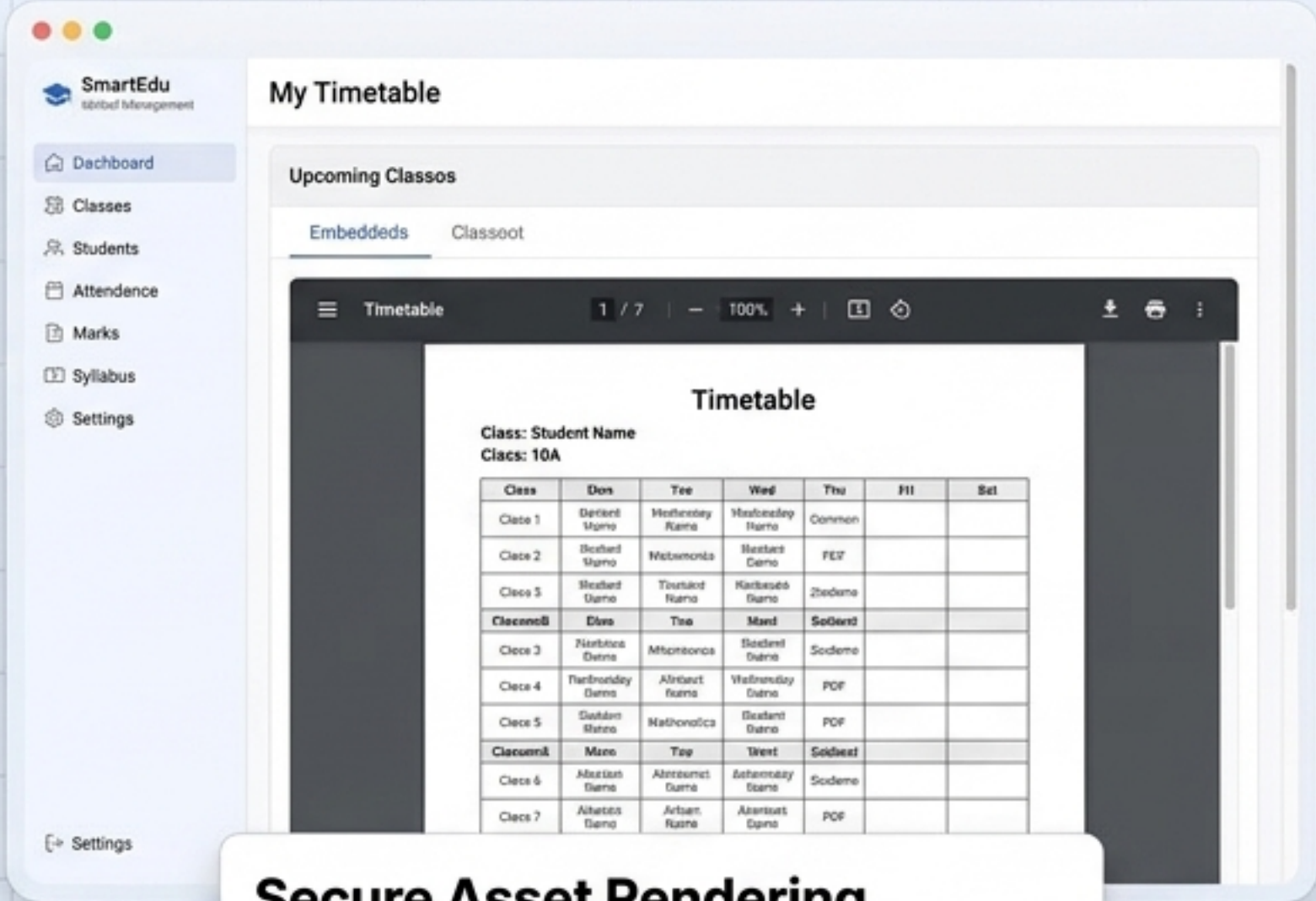
Asset Retrieval: Secure retrieval and rendering of uploaded timetable PDF files directly within the browser tab.

Feature Spotlight: Financial Tracking & Asset Retrieval



Financial Transparency

Tracking paid versus pending tuition with real-time historical transaction logs and visual payment progress indicators.



Secure Asset Rendering

Fetching and embedding raw PDF timetable assets directly into the viewport using backend path retrieval.

Technical Challenges & Engineering Resolutions



❗ Challenge: Unauthorized URL Access

Users attempting to bypass the login screen via direct URL entry.

✅ Resolution: Rigid Session Management

Implemented strict PHP session checks and header redirects to protect restricted pages.

❗ Challenge: Responsive Layout Breaking

Complex data tables breaking visual boundaries on mobile viewports.

✅ Resolution: Tailwind Fluidity

Leveraged Tailwind CSS utility classes (flexbox/grids/overflow-x) to ensure cross-device structural integrity.

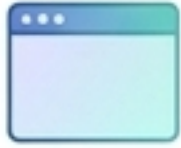
❗ Challenge: Database Query Failures

Connection timeouts and complex JOIN errors during roster loading.

✅ Resolution: Schema & Error Handling

Refined the MySQL relational schema and integrated robust PHP error-handling for stable data pipelines.

Outcomes & Synthesized Competencies



Frontend & UI Design

- HTML5, CSS3, Tailwind CSS, JavaScript.
- Translating data into responsive visual interfaces.



Backend & System Logic

- PHP, User Authentication, State/Session Management, Role-Based Access Control (RBAC).



Database & Tooling

- MySQL, Relational Schema Design, phpMyAdmin, VS Code, GitHub Version Control.



Professional Execution

- Analytical debugging, troubleshooting, team collaboration, and real-time project delivery.

Conclusion: Successfully bridged the gap between isolated academic programming theory and the reality of building scalable, interconnected full-stack software architectures.

Codec Technologies
Internship Certificate